Quick Introduction to Accessible Maths

Science & Engineering Faculty, University of Limerick

Emma Cliffe

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Plan

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- Why worry about maths access?
- What is the maths specific problem?
- What do solutions look like?
- How do you get started?

Live document: https://stem-enable.github.io/Quick-introduction-to-accessible-maths-2026/

Why worry about maths[1] access?

[1] I will use the word maths as a shorthand for all mathematical and statistical content used across any STEM degree. Although I won't be talking about specialist chemical notations and diagrams the same issues apply.

The legal

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Accessibility is a legal requirement... I am not a lawyer[2]. Seek advice.

- Equal access:
  - England, Scotland, Wales: Equality Act 2010
  - Northern Ireland: Disability Discrimination Act 1995, (as amended), including SENDO 2005
  - Ireland: Equal Status Acts 2000 - 2018
- Accessible digital content (including university websites, web services and most VLE content):
  - UK: Public Sector Bodies (Websites and Mobile Applications) (No. 2) Accessibility Regulations 2018
  - Ireland: European Union (Accessibility of Websites and Mobile Applications of Public Sector Bodies) Regulations 2020

[2] Legislation is cited for awareness only and this does not constitute legal advice. Specific legal questions should be referred to your University legal team.

But we should care anyway...

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We want students to spend their effort engaging with concepts, not overcoming barriers in our resources.

Students with print disabilities or difficulties with reading, note-taking, memory or concentration may expend significant effort accessing information before learning can begin.

As educators, we should make information accessible in multiple ways.

- Rich, interactive resources can benefit all students.
- Poorly designed resources create practical and technical barriers:
  - Notation inaccessible via speech, magnification, Braille or small screens
  - Difficult keyboard-only navigation
  - Meaning conveyed by colour alone
  - Poorly scalable graphics

What is the maths specific problem?

Structured text can be unexpectedly inaccessible

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Figure 1: Video demonstration

Could we solve it by...?

Could we solve it by...?

Can we not solve this by ensuring that the PDF is accessible? Unfortunately, the answer is no. The act of creating the PDF destroys the structure of mathematical, and similar, expressions. Reflow on a small screen produces...

$$-b \pm \overline{b^2 - 4ac}$$

$$x\!\!\surd\!= .(1)$$

$$2a$$

What do solutions look like?

Structure + software

Structure + software

Being able to adapt how text is presented, whether for small screens or via assistive software requires:

- Authors to produce accessible materials which maintains the encoding of the structured text
- Readers to have software and assistive technology which can manipulate the specific format

Only four formats of mathematical text are technically accessible:

- HTML using MathJax to render the mathematics
- Word/PowerPoint using the inbuilt equation editor
- Word/PowerPoint using MathType
- EPub3 using MathML

You **must** supply at least one to meet the public bodies requirement.

- Supply your full course content 'spine' in this format.

So, job done?

So, job done?

EHC: SORT OUT FOOTNOTES FOR BEAMER

No. Not all reader assistive technology can access maths all formats!

- Equal access acts mean we must meet individual needs - so may need to produce other formats.
- Some cannot not be transformed easily to others

For authors using Word/PowerPoint, standard accessiblity guidance is not enough.

For authors using the LaTeX scientific document preparation system none of the above formats are native outputs[3]. **And** if a student can't make their own clear print **and print it** you also may need to supply variants on inaccessible PDF!

- Not all accessibility is about technical access
- Clear print[4] PDF is selected most often by disabled students in the Department of Mathematical Sciences at the University of Bath[5].

[3]Work is ongoing to create a version of LaTeX, extra encoding within an

How do you get started?

Word/PowerPoint 365

Word/PowerPoint 365

EHC: CHECK LINKS, CONSIDER WHETHER NEEDS UPDATING

- Use the inbuilt Accessibility Checker and information on e.g. Making your Word documents accessible
- Write *all* mathematical text written using the MS 365 equation editor. E.g. if you are writing about the variable (x) or ( ) it should be written as an equation. If you are writing (x^2) it should be written as an equation.
  - Never use insert symbol.
  - Never write superscripts, subscripts, fractions etc. using font or style changes and standard keyboard input alone
  - Never use an image of an equation.
- Use Review -> Read Aloud (Alt + Ctrl + Space) to check the maths

More information and tools within accessible mathematical Word document workshop resources and effective (keyboard-only) input of maths in Word.

Word/PowerPoint with MathType

Word/PowerPoint with MathType

EHC: CAN THIS BE A NOTE ON THE ABOVE?

Some students will require MathType format

- Because their AT vendor has not implemented the interface to Word but the AT does work with MathType. Test everything and contact the vendors!
- Materials should be prepared as above and then a ***copy*** converted to MathType as automatic conversion of all expressions cannot be reversed.

HTML (WCAG 2.2 AA) $+$ MathJax!

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EHC: THIS NEEDS UPDATING AS EXPERIENCE YOURSELF RELIES ON CHROMEVOX

- Check it meets the legal requirement of WCAG 2.2 Level AA with e.g. Accessibility Insights for Web plugin for Chrome
- Check it is MathJax... Right click!

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

  - Provides/enables: navigation, chunking, zoom, copy/paste, colour, size and layout changes...
  - Structural integrity enables assistive technology including text-to-speech, screenreaders, electronic Braille
  - Fallback for ARIA-aware AT without native support
  - AT providers without native support and not ARIA aware are a problem but not your problem.
- Experience this yourself

Getting there from scientific document preparation systems

Getting there from scientific document preparation systems

EHC: PRESENT OPTIONS WITH LINKS ON ONE SLIDE, UPDATE LINKS

- Experts: tools which may work for a large subset but cannot tell you anything about why something doesn't work
- Non-experts at Bath who want to stick with LaTeX have found that lwarp works for them if they are careful.
  - Beware of some hard work (documentation currently 1237 pages)
  - What happens depends on what you are doing in LaTeX
  - You have to work that out and test for accessibility for yourself.
  - Quick lwarp demo (link for Emma during the talk)

RMarkdown formats

RMarkdown formats

RMarkdown is a language designed for transform to html

- LaTeX for equations
- Simple markup language for the rest of the document.
- Authors are confined to a transformable subset with a quick compile loop and a supportive GUI.
- The output has known accessibility features.
- The typesetting is still extensible, just more reasonably so.
- More information on getting started with RMarkdown see Using R as a basis for writing an accessible mathematical document
- Quick RMarkdown demo (link for Emma during the talk)

Quarto/Bookdown/Clavertondown

Quarto/Bookdown/Clavertondown

EHC ADD QUARTO

Eventually, you will probably want Bookdown:

`{theorem, label="thm1"} Bookdown is needed for things like theorems and internal references`

We built Clavertondown for additional functionality important in pure mathematics and for lecturers generally.

`{newtheorem, env='Thought', label="tho1"} You can create new theorem types without affecting the Bookdown types such as theorem \@ref(thm:thm1).`

"'{newtheorem, env='Nugget', label="nug1"} And you can have theorem types share numbering

"'

Non-text elements

Non-text elements

EHC: DO FIGURES FIRST AND CONSIDER THAT AND ALL BELOW ARE EXTRAS TO LOOK AT AFTER THE TEXT

- Numbas aims to meet WCAG 2.1 AA
- Stack aims to meet WCAG 2.1 AA
- There are probably others but these are the only ones I have used

Simple graphs: Desmos

Simple graphs: Desmos

- Desmos was designed to work and tested with screenreaders and written to conform to WCAG 2.1
- Experience this yourself

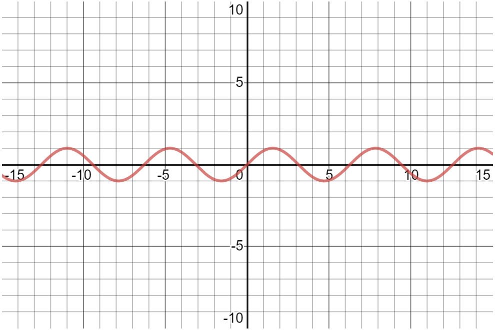

Figure 2: Interactive plot of sine

Geogebra: but be careful!

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- Geogebra aims to meet WCAG 2.1 AA
- But, just because you can make a usable accessible object with a piece of software doesn't mean every object you make is!
- Project **Beyond alt text** between Bath and Coventry produced guidelines
- It is possible and useful but you need to take time and care.

BrailleR: for the statisticians

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EHC: DIRECT TO INSTRUCTIONS?

- BrailleR is a project by and for blind statisticians who use R

Other diagrams

Other diagrams

EHC UPDATE

- DIAGRAM Center has some great resources for those new to diagram description:
  - Poet Image Description Training Tool
  - Image description guidelines
  - Sample book
  - Webinar on complex images
- UKAAF has guidance on accessible images
- NCAM (Old site) has guidelines and examples

Thanks!

Thanks!

Emma Cliffe, E.H.Cliffe@bath.ac.uk

In the UK there is the JISC Accessibility Community Maths Working Group. It may be worth exploring whether links can be made with them from Ireland.